

## Backpropagation

- More info on gradient descent → the derivative tells us how changing the one variable will effect the output, so we can change the variable value accordingly to minimize loss  
or ~~where~~ whatever optimization problems we have

★ Gradient vector → a vector where each value is the derivative of one variable which we are trying to optimize

★  $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) \rightarrow$  chain rule

Basically backward propagation means to make a computation graph, and going backward from the output to get the input values  
→ step by step how we go from input variables to output (usually loss)  
done with differentiation